

 [Learn](#)  [Practice](#)

Hybridisation: domains, geometry and model choice: Construction

Contents and concept route

Concept Book

Hybridisation, electron domains and bonding models

Concept route

1. Hybrid set size
 1. Distinguish the hybrid-set name from the number of orbitals in that set.
2. Count central electron domains
 1. Separate a bonded-atom direction from the number of shared pairs.
 2. The counting centre is nitrogen, not every atom in NF_3 .
 3. Critique the counting operation, rather than merely giving a corrected label.
3. Four domains to sp^3 and tetrahedral geometry
 1. The task asks for a local count before the introductory geometry label.
 2. Map each set name to an orbital inventory, not its superscript alone.

4. Lewis structure to tetrahedral domain arrangement

1. Translate three bonds and one lone pair without adding a fifth direction.

2. Identify whether the error is in connectivity notation or in its interpretation as physical shape.

5. Choose a model by explanatory scope

1. Choose a model by the output requested, rather than the chapter title.

6. Sigma and pi overlap geometry

1. Use the stated internuclear axis to classify the overlap.

7. VSEPR electron-domain versus molecular geometry

1. construct four tetrahedral domain directions and map the three bonds and the lone pair onto them

Hybrid set size

Subject: Chemistry

Topic: Chemical Bonding

Concept: Hybrid set size

Concept difficulty: Medium concept



Distinguish the hybrid-set name from the number of orbitals in that set.

Medium concept

Conditions for this construction

- Can read supplied atoms, single/double/triple bonds, and shared versus lone pairs in a simple Lewis structure.

Why this section matters for the question set

This unit builds toward

- **Q1 · D1**

The move learners miss: Select one valence s and three valence p inputs.

[Try it in Core2](#)

1 Read the requested four equivalent domain directions.

Why valid: The question supplies a tetrahedral four-direction introductory model.

Result: Four model directions must be represented.

2 The step the hard question turns on Select one valence s and three valence p inputs.

Why valid: For a second-period centre, these four independent basis functions can be recombined without changing the basis dimension.

Result: $2s + 2p_x + 2p_y + 2p_z$: four inputs.

3 Name the recombined set and count its members.

Why valid: The name sp^3 records one s and three p inputs; the total number is $1+3$.

Result: sp^3 ; four hybrid orbitals.

4 Compare the count with an sp allocation.

Why valid: One s plus one p has two outputs, so the superscript alone cannot be the set size.

Result: sp has two outputs; the four-member sp^3 inventory is consistent.

Start with direction c

3 directions

Start from the number of directions.

Stage 1 of 3

Show the introductory mapping from 3/4 electron-domain directions to sp^2/sp^3 hybrid set size.

The key step

For second-period atoms, one 2s plus three 2p orbitals gives four sp^3 hybrids. The superscript counts p inputs, not total outputs.

Formula reference • meaning and conditions

Equations and validity

Equation	What it tells you	When you can i
3 electron-domain directions $\leftrightarrow$ $sp^2 \leftrightarrow$ 3 hybrids; 4 directions $\leftrightarrow$ $sp^3 \leftrightarrow$ 4 hybrids	In the introductory valence-bond model, the number of hybrid orbitals equals the number of atomic orbitals combined and matches the modelled electron-domain directions.	Use this as introductory bonding model not as a full quantum description

Worked explanation

An sp^2 set is proposed to contain two hybrids. How many input and output orbitals should it have? Then apply your method to an sp set.

Follow the operation, its justification and its result.

1. A change of orbital basis preserves the number of independent orbitals.
2. Three for sp^2 , two for sp . Repeating the superscript-as-total rule in both cases suggests a held counting rule; correct changed cases after an isolated slip suggest an execution error.
3. For second-period atoms, one $2s$ plus three $2p$ orbitals gives four sp^3 hybrids. The superscript counts p inputs, not total outputs.

Result

Three for sp^2 , two for sp . Repeating the superscript-as-total rule in both cases suggests a held counting rule; correct changed cases after an isolated slip suggest an execution error.

Check

Changed check: an sp set has two hybrids because one s plus one p supplies two basis functions; use this conservation rule on the requested set.

Mistake clinic • diagnose and repair

A tempting wrong path

- The superscript is the total number of hybrid orbitals.

Diagnose

- An sp^2 set is proposed to contain two hybrids. How many input and output orbitals should it have? Then apply your method to an sp set.

Repair

- For second-period atoms, one $2s$ plus three $2p$ orbitals gives four sp^3 hybrids. The superscript counts p inputs, not total outputs.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: an sp set has two hybrids because one s plus one p supplies two basis functions; use this conservation rule on the requested set.

2 • APPLY

Give the input/output inventory for sp without copying the original set.

3 • CONNECT

A change of orbital basis preserves the number of independent orbitals.

Question-specific diagnosis and repair

Repair for Q1

Test the rule on a changed case

An sp² set is proposed to contain two hybrids. How many input and output orbitals should it have? Then apply your method to an sp set.

Your prediction and reason

Now you do one

For a second-period centre allocated an sp set, inventory the hybrid outputs and unused p functions. Show that the valence basis is conserved.

Model answer

Two sp hybrids plus two unused p functions: four outputs from one s plus three p inputs.

1. Two sp hybrids plus two unused p functions: four outputs from one s plus three p inputs.

Practice this concept in Core2

Q1

Sections

Separate a bonded-atom direction from the number of shared pairs.

The counting centre is nitrogen, not every atom in NF_3 .

Critique the counting operation, rather than merely giving a corrected label.

Count central electron domains

Subject: Chemistry

Topic: Chemical Bonding

Concept: Count central electron domains

Concept difficulty: Medium concept

1

Separate a bonded-atom direction from the number of shared pairs.

Medium concept

Section 2 of 11

← Previous

Next →

Conditions for this construction

- Can read supplied atoms, single/double/triple bonds, and shared versus lone pairs in a simple Lewis structure.

Why this section matters for the question set

This unit builds toward

- Q2 · D2

The move learners miss: Count C-to-O as one local VSEPR domain.

[Try it in Core2](#)

1 Make separate neighbour and shared-pair inventories for HCHO.

Why valid: Two H atoms and one O are attached to C; a C=O double bond contains two shared pairs.

Result: Three neighbours; two pairs in C=O.

2 The step the hard question turns on Count C-to-O as one local VSEPR domain.

Why valid: A multiple bond connects one pair of nuclei and supplies one bonded-neighbour direction in local domain counting. Sigma and pi density have different spatial symmetry.

Result: One C=O domain; this does not place pi density on the internuclear axis.

3 Add the two C-H directions.

Why valid: Each of the two distinct C-H connections adds one domain and C has no lone pair.

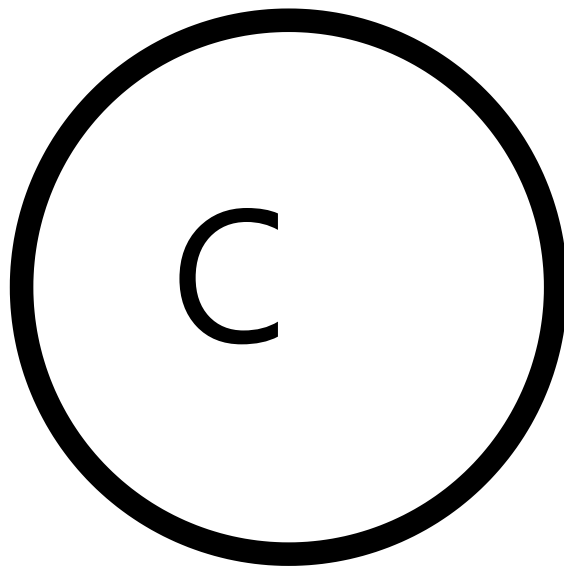
Result: $1+1+1 =$ three C-centred domains.

4 Recount carbon directions in HCN.

Why valid: H-C and C-N connect carbon to two neighbours although C-N has three shared pairs.

Result: Two domains in HCN; shared-pair count and neighbour count are different.

Choose the cou



Mark the atom whose domains are being counted.

Stage 1 of 3

Make local central-atom domain counting visible and show why a multiple bond remains one direction.

The key step

HCHO carbon has three bonded-atom directions. C=O contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.

Formula reference • meaning and conditions

Equations and validity

Equation	What it tells you	When you can
electron-domain count = bonded-neighbour regions + lone-pair regions on the chosen central atom	VSEPR domain counting is local to the selected centre. A single, double or triple bond to one neighbour occupies one region/direction.	• Choose the central atom before counting • Treat each bonded neighbour as one region regardless of bond order

Worked explanation

At carbon in HCN, count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO.

Follow the operation, its justification and its result.

1. The sigma component and pi component of a multiple bond connect the same atom pair.
2. HCN: two carbon directions, three C-N pairs. HCHO: three directions, two C-O pairs.

3. HCHO carbon has three bonded-atom directions. C=O contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.

Result

HCN: two carbon directions, three C-N pairs. HCHO: three directions, two C-O pairs.

Check

Changed check: a triple bond to N in HCN still supplies one C-to-N direction; carbon also has the H direction.

Mistake clinic • diagnose and repair

A tempting wrong path

- Each line of a double bond is a separate local direction.
- Terminal-atom lone pairs add domains at the central atom.
- Bond multiplicity equals neighbour-direction multiplicity.

Diagnose

- At carbon in HCN, count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO.
- Count N-centred domains in NH₃ and NF₃. Then state where the F lone pairs are located.
- A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO.

Repair

- HCHO carbon has three bonded-atom directions. C=O contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.
- NF₃ has three nitrogen-to-fluorine directions and one nitrogen lone pair: four N-centred domains. Terminal F lone pairs do not add N-centred domains.
- Counting both C=O lines as separate domains confuses bond components with one bonded-atom direction. HCHO carbon has three

domains.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: a triple bond to N in HCN still supplies one C-to-N direction; carbon also has the H direction.

2 • APPLY

Explain why replacing a double bond with a triple bond to the same neighbour need not add a direction.

3 • CONNECT

The sigma component and pi component of a multiple bond connect the same atom pair.

2

The counting centre is nitrogen, not every atom in NF₃.

Medium concept

Section 3 of 11

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q8 • D2

The move learners miss: Assign each lone pair to its own atom.

[Try it in Core2](#)

1 Choose nitrogen as the counting centre.

Why valid: The requested domains are centred on N, not on the entire NF₃ molecule.

Result: Three N-F bonding directions are included.

2 The step the hard question turns on Assign each lone pair to its own atom.

Why valid: N has one lone pair; terminal F lone pairs belong to F-centred inventories.

Result: One N lone pair included; terminal F lone pairs excluded from the N count.

3 Add the local entries.

Why valid: The N inventory has three bonded directions plus one central lone pair.

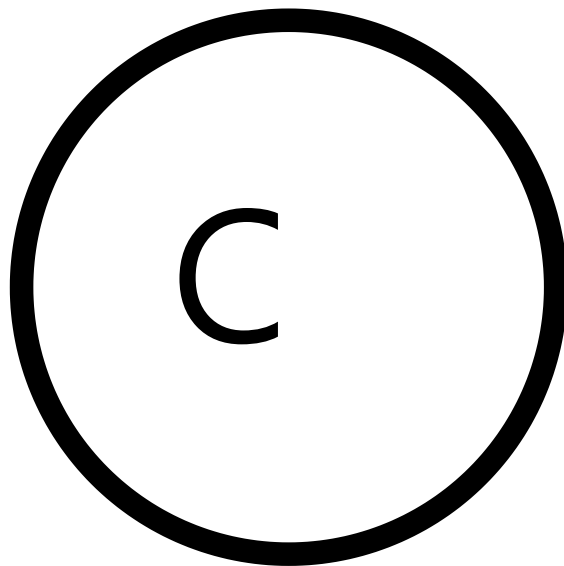
Result: $3+1 =$ four nitrogen-centred domains.

4 Substitute H for F in the local comparison.

Why valid: NH₃ has the same three-bond/one-central-pair N inventory even though terminal atoms differ.

Result: NH₃ also has four N domains; terminal pairs were not required for that total.

Choose the cou



Mark the atom whose domains are being counted.

Stage 1 of 3

Make local central-atom domain counting visible and show why a multiple bond remains one direction.

Worked explanation

Count N-centred domains in NH_3 and NF_3 . Then state where the F lone pairs are located.

Follow the operation, its justification and its result.

1. Local domains consist of bonds and lone pairs at the selected atom.
2. Both N centres have four domains; F pairs are located at F.
3. NF_3 has three nitrogen-to-fluorine directions and one nitrogen lone pair: four N-centred domains. Terminal F lone pairs do not add N-centred domains.

Result

Both N centres have four domains; F pairs are located at F.

Check

Changed check: substituting H for F leaves the local three-bond-plus-lone-pair N inventory unchanged in the introductory model.

Mistake clinic • diagnose and repair

A tempting wrong path

- Each line of a double bond is a separate local direction.
- Terminal-atom lone pairs add domains at the central atom.
- Bond multiplicity equals neighbour-direction multiplicity.

Diagnose

- At carbon in HCN , count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO .
- Count N-centred domains in NH_3 and NF_3 . Then state where the F lone pairs are located.

- A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO.

Repair

- HCHO carbon has three bonded-atom directions. C=O contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.
- NF₃ has three nitrogen-to-fluorine directions and one nitrogen lone pair: four N-centred domains. Terminal F lone pairs do not add N-centred domains.
- Counting both C=O lines as separate domains confuses bond components with one bonded-atom direction. HCHO carbon has three domains.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: substituting H for F leaves the local three-bond-plus-lone-pair N inventory unchanged in the introductory model.

2 • APPLY

Count domains at oxygen in H₂O while excluding electrons assigned to H.

3 • CONNECT

Local domains consist of bonds and lone pairs at the selected atom.

3

Critique the counting operation, rather than merely giving a corrected label.

Medium concept

Section 4 of 11

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- **Q9 · D1**

The move learners miss: Replace bond-line counting with local neighbour

counting. [Try it in Core2](#)

1 Identify the faulty C=O line-counting entry.

Why valid: Two drawn bond lines denote two bond components to one O neighbour.

Result: The student counted one neighbour twice.

2 The step the hard question turns on Replace bond-line counting with local neighbour counting.

Why valid: In VSEPR a multiple bond supplies one bonded-neighbour domain; pi density is lateral rather than an extra atom direction.

Result: C=O contributes one domain, not two.

3 Compute the corrected HCHO total.

Why valid: There are two distinct H neighbours, one O neighbour and no C lone pair.

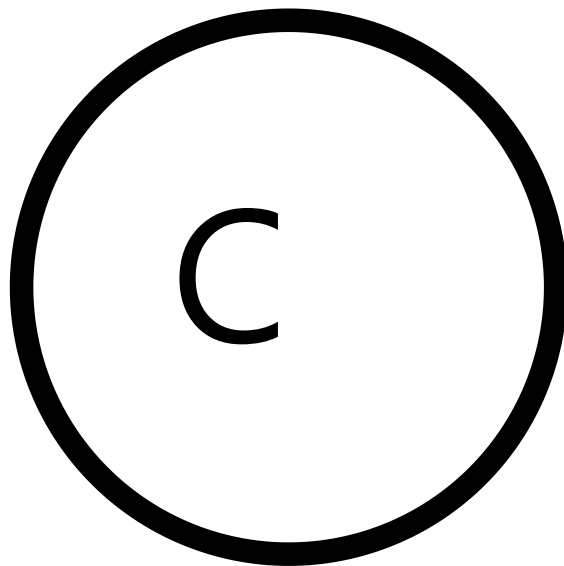
Result: $2+1+0 =$ three C-centred domains.

4 Apply the operation to CO₂.

Why valid: Two C=O bonds connect C to two O neighbours; four drawn bond lines do not imply four local domains.

Result: CO₂ carbon has two domains in the introductory model.

Choose the cou



Mark the atom whose domains are being counted.

Stage 1 of 3

Make local central-atom domain counting visible and show why a multiple bond remains one direction.

Worked explanation

A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO.

Follow the operation, its justification and its result.

1. Bond multiplicity and neighbour directions are different ledgers.
2. HCN two; HCHO three. Repeating line-counting in both cases indicates the rule is held rather than a single arithmetic slip.
3. Counting both C=O lines as separate domains confuses bond components with one bonded-atom direction. HCHO carbon has three domains.

Result

HCN two; HCHO three. Repeating line-counting in both cases indicates the rule is held rather than a single arithmetic slip.

Check

Changed check: HCN carbon has two neighbour directions even though C-N is triple.

Mistake clinic • diagnose and repair

A tempting wrong path

- Each line of a double bond is a separate local direction.
- Terminal-atom lone pairs add domains at the central atom.
- Bond multiplicity equals neighbour-direction multiplicity.

Diagnose

- At carbon in HCN, count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO.

- Count N-centred domains in NH_3 and NF_3 . Then state where the F lone pairs are located.
- A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO.

Repair

- HCHO carbon has three bonded-atom directions. $\text{C}=\text{O}$ contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.
- NF_3 has three nitrogen-to-fluorine directions and one nitrogen lone pair: four N-centred domains. Terminal F lone pairs do not add N-centred domains.
- Counting both $\text{C}=\text{O}$ lines as separate domains confuses bond components with one bonded-atom direction. HCHO carbon has three domains.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: HCN carbon has two neighbour directions even though C-N is triple.

2 • APPLY

Audit a proposed domain count for CO_2 .

3 • CONNECT

Bond multiplicity and neighbour directions are different ledgers.

Question-specific diagnosis and repair

Repair for Q2

Test the rule on a changed case

At carbon in HCN, count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO.

Your prediction and reason

Repair for Q8

Test the rule on a changed case

Count N-centred domains in NH_3 and NF_3 . Then state where the F lone pairs are located.

Your prediction and reason

Repair for Q9

Test the rule on a changed case

A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO.

Your prediction and reason

Now you do one

At carbon in CO_2 , count local domains and then separately count the shared pairs in both C=O bonds. Explain why the totals differ.

Model answer

Two carbon-neighbour domains; four shared pairs. Each double bond connects one O neighbour.

1. Two carbon-neighbour domains; four shared pairs. Each double bond connects one O neighbour.

Practice this concept in Core2

Q2

Q8

Q9

Sections

The task asks for a local count before the introductory geometry label.

Map each set name to an orbital inventory, not its superscript alone.

Four domains to sp^3 and tetrahedral geometry

Subject: Chemistry

Topic: Chemical Bonding

Concept: Four domains to sp^3 and tetrahedral geometry

Concept difficulty: Medium concept

1

The task asks for a local count before the introductory geometry label.

Medium concept

Section 5 of 11

← Previous

Next →

Conditions for this construction

- Can read supplied atoms, single/double/triple bonds, and shared versus lone pairs in a simple Lewis structure.

Why this section matters for the question set

This unit builds toward

- **Q3 · D1**

The move learners miss: Apply the explicitly supplied four-domain rule.

[Try it in Core2](#)

1 Inventory CH₄ at carbon.

Why valid: The supplied Lewis connectivity has four C-H bonds and no C lone pair.

Result: Four bonding domains; zero central lone pairs.

2 **The step the hard question turns on** Apply the explicitly supplied four-domain rule.

Why valid: The task asks for the known introductory correspondence, rather than an experimental determination of orbital coefficients.

Result: Tetrahedral domain geometry; introductory sp³ description.

3 Check the orbital basis ledger.

Why valid: One 2s plus three 2p inputs gives four hybrids, matching four model sigma directions.

Result: Four inputs and four outputs; no residual p in this allocation.

4 Change one substituent without adding a neighbour.

Why valid: CH₃Cl still has four carbon-neighbour connections; different bond properties do not constitute extra domains.

Result: CH₃Cl also has four C-centred domains; no exact bond-property claim follows.

4 C–H bondin

C

Count the four C–H regions around carbon.

Stage 1 of 3

Build methane's four bonding domains into an sp^3 tetrahedral arrangement.

The key step

CH₄ has four bonding domains, no carbon lone pair, and a tetrahedral introductory sp³ description. A Lewis layout encodes connectivity.

Formula reference • meaning and conditions

Equations and validity

Equation	What it tells you	When you can use it
3 electron-domain directions ↔ sp ² ↔ 3 hybrids; 4 directions ↔ sp ³ ↔ 4 hybrids	In the introductory valence-bond model, the number of hybrid orbitals equals the number of atomic orbitals combined and matches the modelled electron-domain directions.	Use this as introductory bonding model not as a covalent quantum description

Worked explanation

Count domains at C in CH₃Cl and C in CH₄. Does redrawing either as a cross change its count?

Follow the operation, its justification and its result.

- Four bonding directions and no central lone pair use the demonstrated four-domain rule.
- Both have four; changing drawing placement changes neither connectivity nor domain count.
- CH₄ has four bonding domains, no carbon lone pair, and a tetrahedral introductory sp³ description. A Lewis layout encodes connectivity.

Result

Both have four; changing drawing placement changes neither connectivity nor domain count.

Check

Changed check: CH₃Cl still has four bonded neighbours around carbon even if the bonds have different properties.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- The page layout is the molecular geometry.
- sp² has two hybrids and sp³ has three.

Diagnose

- Count domains at C in CH₃Cl and C in CH₄. Does redrawing either as a cross change its count?
- Compare proposed totals sp=1, sp²=2, sp³=3 against basis-function conservation. Give the corrected sequence.

Repair

- CH₄ has four bonding domains, no carbon lone pair, and a tetrahedral introductory sp³ description. A Lewis layout encodes connectivity.
- BF₃: three domains and three sp² hybrids. CH₄: four domains and four sp³ hybrids. For these centres use valence 2s/2p language rather than ambiguous 1s + 2p shorthand.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: CH₃Cl still has four bonded neighbours around carbon even if the bonds have different properties.

2 • APPLY

Classify carbon in CH₂Cl₂ by domain inventory, not bond identity.

3 • CONNECT

Four bonding directions and no central lone pair use the demonstrated four-domain rule.

Why this section matters for the question set**This unit builds toward**

- **Q10** · D1

The move learners miss: Match each inventory to a conserved valence-basis allocation. [Try it in Core2](#)

1 Read the two supplied domain inventories.

Why valid: The task supplies three B bonding domains and four C bonding domains, with no central lone pairs.

Result: BF₃: three; CH₄: four.

2 The step the hard question turns on Match each inventory to a conserved valence-basis allocation.

Why valid: One s plus two p yields three hybrids; one s plus three p yields four.

Result: BF₃: sp², three hybrids; CH₄: sp³, four hybrids.

3 Match the ideal domain arrangement within this introductory model.

Why valid: Three equivalent model directions are planar; four are tetrahedral. These labels are model descriptions.

Result: BF₃: trigonal planar; CH₄: tetrahedral.

4 Reconstruct an sp inventory as a third case.

Why valid: One s plus one p provides two inputs and therefore two hybrid outputs.

Result: The set-size sequence is sp=2, sp²=3, sp³=4, rather than the superscripts 1,2,3.

Start with direction c

3 directions

Start from the number of directions.

Stage 1 of 3

Show the introductory mapping from 3/4 electron-domain directions to sp^2/sp^3 hybrid set size.

Worked explanation

Compare proposed totals $sp=1$, $sp^2=2$, $sp^3=3$ against basis-function conservation. Give the corrected sequence.

Follow the operation, its justification and its result.

1. One s plus n p inputs gives $n+1$ hybrid outputs.
2. The corrected sequence is 2,3,4. Repeating the shifted sequence suggests the superscript-as-total rule.
3. BF_3 : three domains and three sp^2 hybrids. CH_4 : four domains and four sp^3 hybrids. For these centres use valence 2s/2p language rather than ambiguous 1s + 2p shorthand.

Result

The corrected sequence is 2,3,4. Repeating the shifted sequence suggests the superscript-as-total rule.

Check

Changed check: sp uses two inputs and two hybrid outputs, providing a third case for the conservation rule.

A tempting wrong path

- The page layout is the molecular geometry.
- sp^2 has two hybrids and sp^3 has three.

Diagnose

- Count domains at C in CH_3Cl and C in CH_4 . Does redrawing either as a cross change its count?
- Compare proposed totals $sp=1$, $sp^2=2$, $sp^3=3$ against basis-function conservation. Give the corrected sequence.

Repair

- CH_4 has four bonding domains, no carbon lone pair, and a tetrahedral introductory sp^3 description. A Lewis layout encodes connectivity.
- BF_3 : three domains and three sp^2 hybrids. CH_4 : four domains and four sp^3 hybrids. For these centres use valence $2s/2p$ language rather than ambiguous $1s + 2p$ shorthand.

1-2-3 QUICK CHECK**1 • CHECK**

Changed check: sp uses two inputs and two hybrid outputs, providing a third case for the conservation rule.

2 • APPLY

Reconstruct all three set sizes without memorising the sequence.

3 • CONNECT

One s plus n p inputs gives $n+1$ hybrid outputs.

Repair for Q3

Test the rule on a changed case

Count domains at C in CH₃Cl and C in CH₄. Does redrawing either as a cross change its count?

Your prediction and reason

Repair for Q10

Test the rule on a changed case

Compare proposed totals $sp=1$, $sp^2=2$, $sp^3=3$ against basis-function conservation. Give the corrected sequence.

Your prediction and reason

Now you do one

Use the introductory local rule for carbon in CH₂Cl₂. Give the domain inventory and model geometry without claiming identical bond properties.

Model answer

Four C-neighbour domains and no lone pair; tetrahedral domain model with introductory sp^3 accounting. Bond identity does not fix lengths or exact angles.

1. Four C-neighbour domains and no lone pair; tetrahedral domain model with introductory sp^3 accounting. Bond identity does not fix lengths or exact angles.

Practice this concept in Core2

Q3

Q10

Sections

Translate three bonds and one lone pair without adding a fifth direction.

Identify whether the error is in connectivity notation or in its interpretation as physical shape.

Lewis structure to tetrahedral domain arrangement

Subject: Chemistry

Topic: Chemical Bonding

Concept: Lewis structure to tetrahedral domain arrangement

Concept difficulty: Medium concept

1

Translate three bonds and one lone pair without adding a fifth direction.

Medium concept

Section 7 of 11

← Previous

Next →

Conditions for this construction

- Can read supplied atoms, single/double/triple bonds, and shared versus lone pairs in a simple Lewis structure.

Why this section matters for the question set

This unit builds toward

- **Q4 · D2**

The move learners miss: Project the four domain directions with a declared depth convention. [Try it in Core2](#)

1 Count exactly three N-H connections and one N lone pair.

Why valid: The question supplies the inventory at nitrogen.

Result: Four domains: three bonds and one lone pair.

2 **The step the hard question turns on** Project the four domain directions with a declared depth convention.

Why valid: Two ordinary lines may denote in-plane directions; a solid wedge denotes toward and a hashed wedge away. The lone-pair slot must replace one bond slot.

Result: N at centre; exactly three labelled H endpoints and one LP direction.

3 Distinguish domain arrangement from nuclear shape.

Why valid: Domain geometry counts all four slots; molecular shape counts the three bonded H nuclei.

Result: Tetrahedral domain model; trigonal-pyramidal molecular shape, without an exact-angle inference.

4 Compare with the four-bond NH_4^+ inventory.

Why valid: Replacing the LP slot with a fourth N-H bond adds a bonded nucleus but no fifth domain.

Result: NH_3 : three H + LP; NH_4^+ : four H. The NH_3 drawing must not contain a fourth H.

NH₃: three N-H bond

Use supplied givens only.

Stage 1 of 3

Construct exactly three N-H domains and one distinguishable lone-pair domain.

The key step

NH₃ has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.

Formula reference • meaning and conditions

—

Equations and validity

Equation	What it tells you	When you can
four electron domains → tetrahedral electron-domain arrangement; molecular geometry omits lone-pair positions from the atom-only name	A flat Lewis structure records connectivity/electron pairs but does not by itself encode the three-dimensional domain geometry.	Four regions around a central atom are arranged tetrahedrally; the introduction of VSEPR model

Worked explanation

Draw NH_3 and a tetrahedral four-bond NH_4^+ comparison. Mark which slot differs and recount the H atoms in each.

Follow the operation, its justification and its result.

1. Wedges and dashes encode depth; electron-domain geometry includes lone pairs, molecular shape uses nuclei.
2. NH_3 has three H plus a lone pair; NH_4^+ has four H and no N lone pair in the local Lewis model.
3. NH_3 has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.

Result

NH_3 has three H plus a lone pair; NH_4^+ has four H and no N lone pair in the local Lewis model.

Check

Count all four slots in the sketch; replacing the lone-pair slot by an N-H bond changes the inventory to an NH_4^+ teaching comparison, not NH_3 .

A tempting wrong path

- Adding depth marks permits an extra bond or domain.
- A Lewis cross proves a square-planar molecule.

Diagnose

- Draw NH_3 and a tetrahedral four-bond NH_4^+ comparison. Mark which slot differs and recount the H atoms in each.
- Does a cross-shaped Lewis CH_4 drawing assert square-planar CH_4 ? Does adding wedge/dash notation change the number of bonds?

Repair

- NH_3 has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.
- For CH_4 a wedge/dash projection represents four tetrahedral bond directions. Projected page angles are not all physical 109.5-degree angles.

1-2-3 QUICK CHECK

1 • CHECK

Count all four slots in the sketch; replacing the lone-pair slot by an N-H bond changes the inventory to an NH_4^+ teaching comparison, not NH_3 .

2 • APPLY

Create an H_2O domain sketch with two bonded slots and two lone-pair slots.

3 • CONNECT

Wedges and dashes encode depth; electron-domain geometry includes lone pairs, molecular shape uses nuclei.

2

Identify whether the error is in connectivity notation or in its interpretation as physical shape.

Medium concept

This unit builds toward

- **Q7 · D2**

The move learners miss: Declare and use a wedge/dash convention.

[Try it in Core2](#)

1 Retain the four C-H connections from Lewis notation.

Why valid: A Lewis cross records connectivity; its page angles do not measure molecular angles.

Result: C at centre with four H endpoints; no bond is added or removed.

2 The step the hard question turns on Declare and use a wedge/dash convention.

Why valid: A solid wedge points toward the viewer, a hashed wedge away; ordinary lines lie in the chosen page plane.

Result: Two in-plane bonds, one toward bond and one away bond represent a tetrahedral model.

3 Explain the physical and projected angles separately.

Why valid: The ideal tetrahedral model has about 109.5-degree physical angles, while a projection distorts page angles.

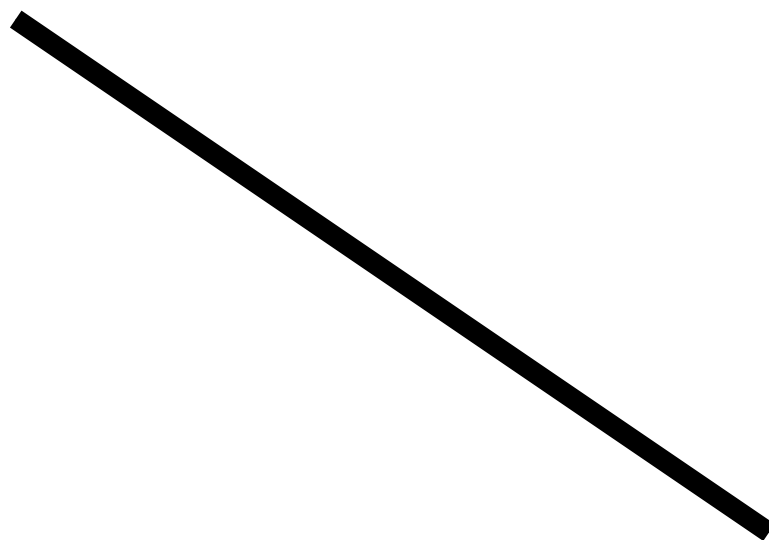
Result: The flat cross does not assert square-planar CH₄; depth belongs to the 3D interpretation.

4 Change viewpoint and recount endpoints.

Why valid: A rotation or a different projection preserves connectivity and noncoplanar geometry.

Result: Four C-H bonds remain; the drawing is a projection of the same tetrahedron.

H



C

Keep carbon bonded to four hydrogens.

Stage 1 of 3

Use wedge/dash depth encoding to distinguish tetrahedral methane from a flat cross Lewis drawing.

Worked explanation

Does a cross-shaped Lewis CH₄ drawing assert square-planar CH₄? Does adding wedge/dash notation change the number of bonds?

Follow the operation, its justification and its result.

1. A flat Lewis drawing does not assert that all nuclei are coplanar.
2. No and no.
3. For CH₄ a wedge/dash projection represents four tetrahedral bond

directions. Projected page angles are not all physical 109.5-degree angles.

Result

No and no.

Check

Changed check: rotate the paper or choose another viewing direction; the atom connectivity and noncoplanar model must stay unchanged.

Mistake clinic • diagnose and repair

A tempting wrong path

- Adding depth marks permits an extra bond or domain.
- A Lewis cross proves a square-planar molecule.

Diagnose

- Draw NH_3 and a tetrahedral four-bond NH_4^+ comparison. Mark which slot differs and recount the H atoms in each.
- Does a cross-shaped Lewis CH_4 drawing assert square-planar CH_4 ? Does adding wedge/dash notation change the number of bonds?

Repair

- NH_3 has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.
- For CH_4 a wedge/dash projection represents four tetrahedral bond directions. Projected page angles are not all physical 109.5-degree angles.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: rotate the paper or choose another viewing direction; the atom connectivity and noncoplanar model must stay unchanged.

2 • APPLY

Explain two different projections of the same tetrahedral model.

3 • CONNECT

A flat Lewis drawing does not assert that all nuclei are coplanar.

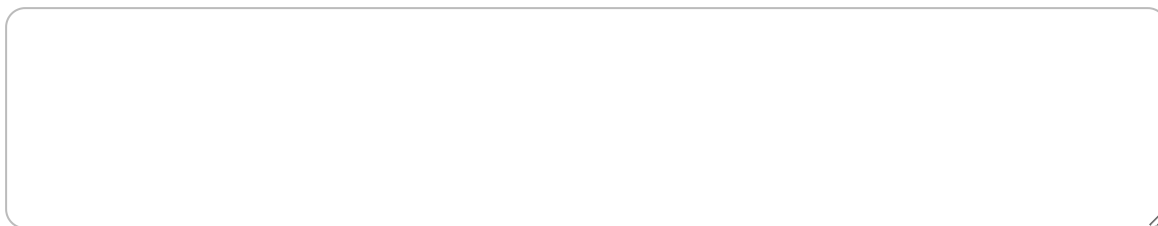
Question-specific diagnosis and repair

Repair for Q4

Test the rule on a changed case

Draw NH_3 and a tetrahedral four-bond NH_4^+ comparison. Mark which slot differs and recount the H atoms in each.

Your prediction and reason

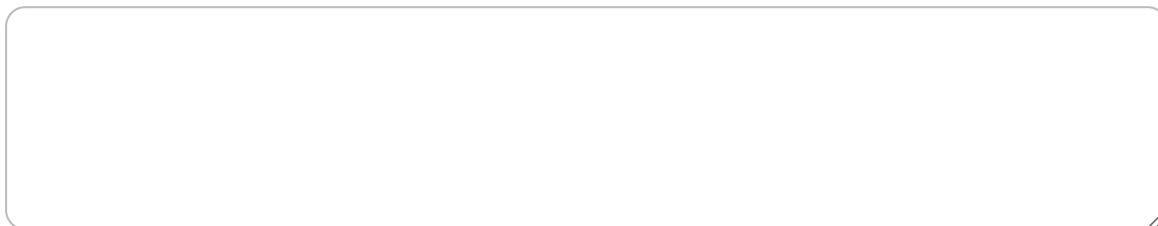


Repair for Q7

Test the rule on a changed case

Does a cross-shaped Lewis CH_4 drawing assert square-planar CH_4 ? Does adding wedge/dash notation change the number of bonds?

Your prediction and reason



Now you do one

Draw a depth-labelled domain model for H_2O : two O-H bonds and two central lone pairs. Separate domain arrangement from bonded-nucleus shape.

Model answer

Four O-centred domains: two bonds and two lone pairs. Introductory tetrahedral domain arrangement; bent molecular shape. Exactly two H endpoints.

1. Four O-centred domains: two bonds and two lone pairs. Introductory tetrahedral domain arrangement; bent molecular shape. Exactly two H endpoints.

Practice this concept in Core2

Q4

Q7

Choose a model by explanatory scope

Subject: Chemistry

Topic: Chemical Bonding

Concept: Choose a model by explanatory scope

Concept difficulty: Medium concept

1

Choose a model by the output requested, rather than the chapter title.

Medium concept

Section 9 of 11

← Previous

Next →

Conditions for this construction

- Can read supplied atoms, single/double/triple bonds, and shared versus lone pairs in a simple Lewis structure.

Why this section matters for the question set

—

This unit builds toward

- **Q5** · D2 · the hardest question in this set
The move learners miss: Choose VSEPR for that output. [Try it in Core2](#)

1 Specify the required output before choosing a model.

Why valid: The task asks for qualitative domain arrangement and expressly excludes detailed orbital overlap.

Result: Required output: qualitative three-dimensional domain geometry.

2 The step the hard question turns on Choose VSEPR for that output.

Why valid: VSEPR uses the three bonding domains and central lone pair to describe their approximate spatial arrangement.

Result: Four-domain, approximately tetrahedral arrangement; pyramidal bonded-nucleus shape.

3 Declare unresolved quantities.

Why valid: A qualitative domain model does not supply orbital wavefunctions, overlap amplitudes or exact measured angles.

Result: No detailed orbital composition, exact angle or spectrum is determined.

4 Change the required output to axial versus lateral overlap.

Why valid: That question needs orbital orientation/symmetry information beyond a geometry-only model.

Result: An orbital-overlap account is now required; familiarity alone cannot select the model.

Q5 asks for overall 3D electr

Identify that Q5 asks for overall 3D electron-domain arrangement.

Stage 1 of 3

Compare what VSEPR and orbital-overlap descriptions predict for the same NH_3 anchor.

The key step

VSEPR fits qualitative electron-domain arrangement. An orbital-overlap model addresses orbital participation. VSEPR does not itself calculate an exact experimental angle or full electronic structure.

Formula reference · meaning and conditions

Equations and validity

Equation	What it tells you	When you can
model choice follows requested explanatory output	VSEPR is the economical model for overall electron-domain arrangement; orbital-overlap/valence-bond descriptions answer orbital composition and overlap questions.	Choose the model after identifying the question to predict

Worked explanation

Task A asks for atom/domain arrangement; task B asks which orbitals overlap along the bond axis. State what information each model must supply.

Follow the operation, its justification and its result.

1. A model has a defined prediction scope and unresolved questions.
2. A needs a geometry model; B needs an orbital account. Choosing the same familiar model without checking output in both cases suggests scope confusion.
3. VSEPR fits qualitative electron-domain arrangement. An orbital-overlap model addresses orbital participation. VSEPR does not itself calculate an exact experimental angle or full electronic structure.

Result

A needs a geometry model; B needs an orbital account. Choosing the same familiar model without checking output in both cases suggests scope confusion.

Check

Changed task: detailed axial versus side-on overlap now requires an overlap description; a geometry-only account is insufficient.

Mistake clinic • diagnose and repair

A tempting wrong path

- A familiar chapter keyword selects the model.

Diagnose

- Task A asks for atom/domain arrangement; task B asks which orbitals overlap along the bond axis. State what information each model must supply.

Repair

- VSEPR fits qualitative electron-domain arrangement. An orbital-overlap model addresses orbital participation. VSEPR does not itself calculate an exact experimental angle or full electronic structure.

1-2-3 QUICK CHECK

1 • CHECK

Changed task: detailed axial versus side-on overlap now requires an overlap description; a geometry-only account is insufficient.

2 • APPLY

Choose the needed model when the new task asks for pi-overlap orientation in ethene.

3 • CONNECT

A model has a defined prediction scope and unresolved questions.

Question-specific diagnosis and repair

—

Repair for Q5

Test the rule on a changed case

Task A asks for atom/domain arrangement; task B asks which orbitals overlap along the bond axis. State what information each model must supply.

Your prediction and reason

Now you do one

A new task asks which orbital directions could produce the pi component in ethene and which plane contains its node. Choose the information needed and name one prediction a domain-count-only account cannot supply.

Model answer

An orbital-overlap account is needed: perpendicular p functions overlap laterally with a node in the molecular plane. Domain counting alone does not supply this orbital symmetry.

1. An orbital-overlap account is needed: perpendicular p functions overlap laterally with a node in the molecular plane. Domain counting alone does not supply this orbital symmetry.

Practice this concept in Core2

Q5

Sigma and pi overlap geometry

1

Use the stated internuclear axis to classify the overlap.

Medium concept

Section 10 of 11

← Previous

Next →

Conditions for this construction

- Can read supplied atoms, single/double/triple bonds, and shared versus lone pairs in a simple Lewis structure.

Why this section matters for the question set**This unit builds toward**

- Q6 · D1

The move learners miss: Classify A from its stated end-on overlap.

[Try it in Core2](#)

1 Mark the line joining the nuclei in each sketch.

Why valid: Sigma/pi labels refer to overlap relative to this internuclear axis, not to page orientation.

Result: A common geometric reference is declared.

2 The step the hard question turns on Classify A from its stated end-on overlap.

Why valid: The supplied overlap is axial; in the introductory sketch the sigma component is symmetric about the bond axis.

Result: A is sigma overlap.

3 Classify B from side-on p overlap.

Why valid: The supplied parallel lateral lobes have a nodal plane containing the axis.

Result: B is pi overlap.

4 Rotate the whole picture, including its axis.

Why valid: A rigid rotation preserves all relative orientation and symmetry relations.

Result: Neither classification changes under whole-picture rotation.

internuclear axis

Use the atom-atom line as the reference.

Stage 1 of 3

Anchor sigma/pi classification to the internuclear axis and overlap orientation.

The key step

Sigma overlap is end-on/axial in the supplied sketch; pi overlap is lateral with a nodal plane containing the axis.

Formula reference • meaning and conditions

—

Equations and validity

Equation	What it tells you	When you use it
sigma: end-on overlap along internuclear axis; pi: side-on overlap with density on opposite sides of the axis	Sigma/pi classification is defined by overlap symmetry/orientation relative to the internuclear axis, not merely by seeing a single or double bond label.	• Use internuclear axis as reference direction

Worked explanation

Rotate the whole end-on sketch by 90 degrees on paper. Does it become pi? Compare that with rotating one p orbital relative to its neighbour.

Follow the operation, its justification and its result.

1. End-on and side-on are orientation/symmetry descriptions, not molecule-name clues.
2. Whole-picture rotation does not change sigma/pi classification; relative orbital rotation can change overlap. Conflating page orientation with relative geometry suggests the misconception.
3. Sigma overlap is end-on/axial in the supplied sketch; pi overlap is lateral with a nodal plane containing the axis.

Result

Whole-picture rotation does not change sigma/pi classification; relative orbital rotation can change overlap. Conflating page orientation with relative geometry suggests the misconception.

Check

Changed check: rotating the entire sketch on the page leaves its classification unchanged relative to the rotated axis.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- Turning a drawing on the page changes sigma into pi.

Diagnose

- Rotate the whole end-on sketch by 90 degrees on paper. Does it become pi? Compare that with rotating one p orbital relative to its neighbour.

Repair

- Sigma overlap is end-on/axial in the supplied sketch; pi overlap is lateral with a nodal plane containing the axis.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: rotating the entire sketch on the page leaves its classification unchanged relative to the rotated axis.

2 • APPLY

Label a sigma component within a triple bond using its geometry.

3 • CONNECT

End-on and side-on are orientation/symmetry descriptions, not molecule-name clues.

Question-specific diagnosis and repair

Repair for Q6

Test the rule on a changed case

Rotate the whole end-on sketch by 90 degrees on paper. Does it become pi? Compare that with rotating one p orbital relative to its neighbour.
Your prediction and reason

Now you do one

An end-on sketch is rotated together with its internuclear axis by 45 degrees on the page. Classify the overlap and justify the invariant you used.

Model answer

It remains sigma: relative axial overlap is unchanged by rigid rotation.

1. It remains sigma: relative axial overlap is unchanged by rigid rotation.

VSEPR electron-domain versus molecular geometry

Subject: Chemistry

Topic: Chemical Bonding

Concept: VSEPR electron-domain versus molecular geometry

Concept difficulty: Medium concept

1

construct four tetrahedral domain directions and map the three bonds and the lone pair onto them

Medium concept

Section 11 of 11

← Previous

Conditions for this construction

- Can read supplied atoms, single/double/triple bonds, and shared versus lone pairs in a simple Lewis structure.

1 Count exactly three N-H connections and one N lone pair.

Why valid: The question supplies the inventory at nitrogen.

Result: Four domains: three bonds and one lone pair.

2 Project the four domain directions with a declared depth convention.

Why valid: Two ordinary lines may denote in-plane directions; a solid wedge denotes toward and a hashed wedge away. The lone-pair slot must replace one bond slot.

Result: N at centre; exactly three labelled H endpoints and one LP direction.

3 Distinguish domain arrangement from nuclear shape.

Why valid: Domain geometry counts all four slots; molecular shape counts the three bonded H nuclei.

Result: Tetrahedral domain model; trigonal-pyramidal molecular shape, without an exact-angle inference.

4 Compare with the four-bond NH_4^+ inventory.

Why valid: Replacing the LP slot with a fourth N-H bond adds a bonded nucleus but no fifth domain.

Result: NH_3 : three H + LP; NH_4^+ : four H. The NH_3 drawing must not contain a fourth H.

Count all four electron



Count all four electron domains around nitrogen.

Stage 1 of 3

Distinguish the electron-domain geometry and atom-only molecular geometry predicted by VSEPR for NH_3 .

The key step

NH_3 has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.

Formula reference • meaning and conditions

Equations and validity

Equation	What it tells you	When you can
four electron domains $\rightarrow$ tetrahedral electron-	A flat Lewis structure records connectivity/electron	Four regic around a

Equation	What it tells you	When you can
domain arrangement; molecular geometry omits lone-pair positions from the atom-only name	pairs but does not by itself encode the three- dimensional domain geometry.	are arrang tetrahedr the introd VSEPR mc

Worked explanation

Draw NH_3 and a tetrahedral four-bond NH_4^+ comparison. Mark which slot differs and recount the H atoms in each.

Follow the operation, its justification and its result.

1. Wedges and dashes encode depth; electron-domain geometry includes lone pairs, molecular shape uses nuclei.
2. NH_3 has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.

Result

NH_3 has three H plus a lone pair; NH_4^+ has four H and no N lone pair in the local Lewis model. A recurring fourth H/fifth domain exposes representation substitution.

Check

Count all four slots in the sketch; replacing the lone-pair slot by an N-H bond changes the inventory to an NH_4^+ teaching comparison, not NH_3 .

Mistake clinic • diagnose and repair

—

A tempting wrong path

- Copying the planar Lewis sketch as the molecular shape, or putting all four domains in one plane.

Diagnose

- the invariant three N–H bonding domains plus one nitrogen lone-pair domain while moving from 2D Lewis notation to a 3D domain picture

Repair

- Preserve the four-domain count first, then map those domains to tetrahedral directions; molecular geometry names atom positions and therefore omits the lone-pair direction.

1-2-3 QUICK CHECK

1 • CHECK

CHECK: State the relevant invariant without using the final answer.

2 • APPLY

APPLY: Use the same rule on a nearby molecule or representation.

3 • CONNECT

CONNECT: Explain which representation/model level the rule belongs to.

Now you do one

Draw a depth-labelled domain model for H₂O: two O–H bonds and two central lone pairs. Separate domain arrangement from bonded-nucleus shape.

Model answer

Four O-centred domains: two bonds and two lone pairs. Introductory tetrahedral domain arrangement; bent molecular shape. Exactly two H endpoints.

- Four O-centred domains: two bonds and two lone pairs. Introductory tetrahedral domain arrangement; bent molecular shape. Exactly two H endpoints.